SIMATS ENGINEERING
ASSESSMENT TOOL2

COURSE CODE: MLA 02 **COURSE NAME: Fundamentals Of Machine Learning**

COURSE OUTCOME COVERED

CO2: *Apply supervised learning algorithms for regression, classification, and predictive modeling tasks to solve structured data problems in practical applications.* *(BL3)*

Assessment Tool Weightage: 20%

QUESTIONS: 10 **Total Marks:50**

Annexure A

Scenario-Based

Code of Conduct:

I ***R.Pranathi(192525076)*** *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: R. Pranathi

Annexure A

Scenario-Based

Q1	Apply a linear regression model to predict apartment rent based on floor area, number of rooms, and location in a real-estate scenario.	5 Marks	6 min
Q2	Use a linear classification model to categorize emails as spam or non-spam using word frequency features in a mail service system.	5 Marks	6 min
Q3	Apply the Naïve Bayes algorithm to predict whether a patient has flu based on symptoms such as fever, cough, and headache.	5 Marks	6 min
Q4	Use discriminant functions to classify loan applicants into eligible and non-eligible groups based on income and credit score.	5 Marks	6 min
Q5	Apply a probabilistic generative model to classify handwritten digits in an optical character recognition system.	5 Marks	6 min
Q6	Implement a probabilistic discriminative model to predict customer churn from service usage data in a telecom scenario.	5 Marks	6 min
Q7	Apply Bayesian logistic regression to predict whether a customer will purchase a product after viewing online advertisements.	5 Marks	6 min
Q8	Use linear regression to forecast monthly sales for a retail company based on advertising expenditure and seasonal demand.	5 Marks	6 min
Q9	Apply Naïve Bayes classification in a news portal to categorize articles into sports, politics, and entertainment.	5 Marks	6 min
Q10	Use Bayesian logistic regression to predict student placement outcomes based on CGPA, coding skills, and aptitude scores.	5 Marks	6 min

RUBRICS FOR CALCULATION

Scenario based assessment

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

1ANS:

1. Introduction

Linear regression is a supervised machine learning algorithm used to predict continuous numerical values. In this scenario, it helps estimate the monthly rent of an apartment using its features.

2. Identify the input and output variables

- Floor area in square feet.
- Number of rooms.
- Location of the apartment.
- Output: Predicted monthly rent.

Location can be represented using numerical variables, such as separate binary indicators for different areas.

3. Construct the linear regression model

The multiple linear regression equation is:

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3$$

Where:

- Y = Predicted monthly rent.
- x_1 = Floor area.
- x_2= Number of rooms.
- x_3= Encoded location features.
- β_0= Intercept.
- β1,β2,β3 = Regression coefficients.

4. Implementation

First, collect historical apartment data containing floor area, room count, location, and actual rent. Preprocess the data and convert location into numerical features. Train the model using the historical data and evaluate its performance using metrics such as Mean Absolute Error (MAE).

5. Decision and conclusion

The trained model predicts the monthly rent of a new apartment. The company can use this prediction to estimate suitable rental prices.

Result: Linear regression provides a systematic method for predicting apartment rent from area, rooms, and location.

2ANS:

1. Introduction

Linear classification is a supervised learning technique that separates data into classes using a linear decision boundary. It can classify emails based on the frequency of suspicious words.

2. Identify the input and output variables

Input features include:

- Frequency of the word "offer".
- Frequency of the word "win".
- Frequency of suspicious links or other selected words.

Output classes:

- Spam = 1.
- Non-spam = 0.

3. Construct the classification model

The linear classification function is:

$$Z = \omega_0 - \omega_1 x_1 + \omega_2 x_2$$

Where:

- x_1= Frequency of "offer".
- x_2= Frequency of "win".
- w_0, w_1, ω_2 = Model parameters.

The prediction rule is:

$$\hat{y} = \begin{cases} 1, & z \geq 0 \\ 0, & z < 0 \end{cases}$$

4. Implementation

Collect a dataset of labelled emails. Extract word frequencies and split the dataset into training and testing sets. Train a linear classifier, such as a perceptron or logistic regression classifier, and evaluate its accuracy.

5. Decision and conclusion

If an incoming email is classified as spam, the system can move it to the spam folder. Otherwise, it can deliver the email to the inbox.

Result: Linear classification helps an email service automatically identify spam messages using word frequency features.

3ANS:

1. Introduction

Naïve Bayes is a probabilistic supervised learning algorithm based on Bayes' theorem. It predicts a class by calculating its probability using the observed features.

2. Identify the input and output variables

Input features:

- Fever.
- Cough.
- Headache.

Output classes:

- Flu = Yes.
- Flu = No.

3. Apply Bayes' theorem

The formula is:

$$(\text{Flu} \mid \text{Symptoms}) = \frac{\text{P(Symptoms|Flu) P(Flu)}}{\text{P(Symptoms)}}$$

Under the Naïve Bayes assumption, the symptoms are conditionally independent given the flu class.

Therefore:

$$P(\text{Flu} \mid F,C,H) \propto P(\text{Flu})\ P(F \mid \text{Flu})P(C \mid \text{Flu})P(H \mid \text{Flu})$$

Where:

- F = Fever.
- C = Cough.
- H = Headache.

4. Implementation

Collect historical patient records with symptoms and confirmed flu status. Calculate prior probabilities and conditional probabilities from the training data. For a new patient, calculate the posterior probability of both classes and select the class with the higher probability.

Laplace smoothing can be used to avoid zero probabilities when a symptom combination is absent from the training dataset.

5. Decision and conclusion

If the estimated probability of flu is higher than the probability of no flu, the model predicts that the patient may have flu. This prediction should support, not replace, professional medical diagnosis.

Result: Naïve Bayes can classify patients based on symptoms and estimate the probability of flu.

4ANS:

1. Introduction

Discriminant analysis is a supervised classification technique that finds a function for distinguishing between different classes. A bank can use it to assist in evaluating loan applicants.

2. Identify the input and output variables

Input features:

- Applicant's annual income.
- Credit score.

Output classes:

- Eligible.
- Non-eligible.

3. Construct the discriminant function

A linear discriminant function can be represented as:

$$g(X) = w_0 + w_1 x_1 + \omega_2 x_2$$

Where:

- x_1= Annual income.
- x_2= Credit score.
- $\omega_0, \omega_1, \omega_2$= Model parameters.

The classification rule is:

$$\hat{y} = \begin{cases} Eligible & g(x) \geq 0 \\ Non-eligible, & g(x) < 0 \end{cases}$$

The coefficients and threshold must be estimated from labelled applicant data.

4. Implementation

Collect historical loan applications containing income, credit scores, and loan eligibility decisions. Train a discriminant analysis model using these records. Evaluate the model on unseen data to determine how accurately it classifies applicants.

5. Decision and conclusion

The model can help a bank identify applicants who meet the patterns associated with previous eligible applicants. However, actual loan decisions should also consider applicable lending rules and other relevant factors.

Result: Discriminant analysis provides a mathematical method for classifying loan applicants based on income and credit score.

5ANS:

1. Introduction

A probabilistic generative model learns how the input features are distributed for each class. It then predicts the class that most probably generated the observed input.

For handwritten digit recognition, a generative model can learn the patterns associated with each digit.

2. Identify the input and output variables

Input:

- Pixel values of a handwritten digit image.

Output:

- One of the ten classes: 0, 1, 2, 3, 4, 5, 6, 7, 8, or 9.

3. Apply the probabilistic model

Using Bayes' theorem:

$$P(Y|X) = \frac{P(X|Y)P(Y)}{P(X)}$$

Where:

- X = Pixel features of the image.
- Y = Digit class.
- P(Y) = Prior probability of the digit.
- P(X|Y) = Likelihood of observing the image features for that digit.
- P(Y|X) = Posterior probability of the digit.

The predicted digit is:

$$\hat{y} = arg\ \max_{y} P(x|y)P(y)$$

4. Implementation

Collect labelled images of handwritten digits. Preprocess the images by resizing them and normalizing pixel values. Train a generative model, such as a Gaussian Naïve Bayes classifier using suitable image features, or a generative model designed for image data.

Test the model on unseen handwritten images.

5. Decision and conclusion

The system calculates the probability of each digit class and selects the most probable digit. The predicted digit can then be converted into text.

Result: A probabilistic generative model can support automatic handwritten digit recognition in OCR systems.

6ANS:

1. Introduction

A probabilistic discriminative model directly estimates the probability of a class given the input features. Logistic regression is a commonly used probabilistic discriminative model for binary classification.

2. Identify the input and output variables

Input features may include:

- Monthly service charges.
- Number of months the customer has used the service.
- Number of complaints.
- Frequency of service usage.

Output:

- Churn = 1.
- No churn = 0.

3. Construct the logistic regression model

The logistic regression equation is:

$$P(Y=1 \mid X) = \frac{1}{1+e^{-z}}$$

Where:

$$Z = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n$$

The model estimates the probability that a customer will leave the service.

4. Implementation

Collect historical customer records with service usage information and churn status. Clean the dataset, encode categorical features, and divide the data into training and testing sets. Train the logistic regression model and evaluate it using accuracy, precision, recall, and other appropriate metrics.

5. Decision and conclusion

If the predicted churn probability is at least 0.5, the customer can be classified as likely to churn under the chosen threshold. The company can then consider appropriate retention measures.

Result: Logistic regression helps telecom companies identify customers who may discontinue their services.

7ANS:

1. Introduction

Bayesian logistic regression combines logistic regression with Bayesian inference. It estimates purchase probability while incorporating prior distributions over the model parameters.

2. Identify the input and output variables

Input features:

- Number of advertisements viewed.
- Time spent on the product page.
- Previous purchases.
- Customer engagement with advertisements.

Output:

- Purchase = 1.
- No purchase = 0.

3. Formulate the model

The logistic regression equation is:

$$P(Y=1|X,\beta) = \frac{1}{1+e^{-\beta^T X}}$$

Where:

- X = Customer features.
- β = Regression coefficients.
- Y=1 indicates that the customer purchases the product.

In Bayesian logistic regression, prior distributions are assigned to the coefficients:

$$\beta_j \sim N(0, \sigma^2)$$

The posterior distribution is:

$$P(\beta|D) \propto P(D|\beta)P(\beta)$$

Where D represents the observed customer data.

4. Implementation

Collect historical advertising and purchase records. Preprocess the features and train a Bayesian logistic regression model using an appropriate inference method. Evaluate its predictions on unseen customers.

5. Decision and conclusion

The model estimates the probability that a customer will purchase the product. The company can use this information to improve advertisement targeting and marketing decisions.

Result: Bayesian logistic regression provides purchase probability estimates while accounting for uncertainty in the model parameters.

8ANS:

1. Introduction

Linear regression predicts continuous numerical values by modelling the relationship between a dependent variable and one or more independent variables. It can be used to estimate future retail sales from advertising expenditure and seasonal factors.

2. Identify the input and output variables

Input features:

- Advertising expenditure.
- Seasonal demand indicator.

Output:

- Monthly sales revenue.

3. Construct the regression model

The multiple linear regression equation is:

$$Y = \beta_0 + B_1 x_1 + \beta_2 x_2$$

Where:

- Y = Monthly sales.
- x_1 = Advertising expenditure.
- x_2 = Seasonal demand.
- β_0= Intercept.
- β_1, β_2= Regression coefficients.

If seasonal demand is represented by categories, such as summer, winter, and festival season, appropriate indicator variables can be used.

4. Implementation

Collect historical monthly sales, advertising expenditure, and seasonal information. Train the model using past data and evaluate its performance using MAE or Root Mean Squared Error (RMSE).

For forecasting, provide the planned advertising expenditure and relevant seasonal features for the future month.

5. Decision and conclusion

The predicted sales can help the company plan inventory, allocate advertising budgets, and prepare for periods of higher or lower demand.

Result: Linear regression can forecast monthly retail sales using advertising expenditure and seasonal demand information.

9ANS:

1. Introduction

Naïve Bayes is a probabilistic classification algorithm based on Bayes' theorem. It is widely applicable to text classification because it can estimate the probability of a category from the words appearing in an article.

2. Identify the input and output variables

Input:

- Words and their frequencies in a news article.

Output classes:

- Sports.
- Politics.
- Entertainment.

3. Apply Bayes' theorem

The posterior probability of a category is:

$$P(C|X) = \frac{P(X|C)P(C)}{P(X)}$$

Where:

- C = News category.
- X = Words in the article.
- P(C) = Prior probability of the category.
- P(X|C) = Likelihood of the article's words given the category.

The predicted category is:

$$\hat{C} = arg\ \underset{c}{max} P(C|X)$$

For a simple Multinomial Naïve Bayes model, the word frequencies are used to estimate the likelihood of each category.

4. Implementation

Collect news articles labelled as sports, politics, or entertainment. Convert the text into numerical features using a Bag-of-Words or TF-IDF representation. Train a Multinomial Naïve Bayes classifier and test it on unseen articles.

5. Decision and conclusion

The model calculates the probability of each category and assigns the article to the category with the highest estimated probability.

For example, an article containing words frequently associated with cricket and football may be classified as sports.

Result: Naïve Bayes can automatically categorize news articles and help users find relevant news more easily.

10ANS:

1. Introduction

Bayesian logistic regression is a probabilistic classification technique that combines logistic regression with Bayesian inference. It can estimate the probability of a student being placed while representing uncertainty in the regression coefficients.

2. Identify the input and output variables

Input features:

- CGPA.
- Coding skills score.
- Aptitude test score.

Output:

- Placed = 1.
- Not placed = 0.

3. Formulate the model

The logistic regression equation is:

$$P(Y=1|X) = \frac{1}{1+e^{-z}}$$

Where:

$$Z = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3$$

Here:

- x_1= CGPA.
- x_2= Coding skills score.
- x_3= Aptitude score.
- Y = 1 indicates placement.

Bayesian logistic regression assigns prior distributions to the coefficients and estimates their posterior distributions using historical placement data.

4. Implementation

Collect historical student data containing CGPA, coding skills, aptitude scores, and placement outcomes. Clean and preprocess the dataset, then train the Bayesian logistic regression model.

Evaluate the model using accuracy, precision, recall, and a confusion matrix.

5. Decision and conclusion

For a new student, the model estimates the probability of placement based on the student's features. A probability threshold can be used to classify the predicted outcome.

The institution can use these predictions to identify students who may benefit from additional coding practice, aptitude training, or placement preparation.

Result: Bayesian logistic regression can help estimate student placement probabilities and support targeted career guidance.